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Research and Applications

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Designing interactive visualizations for analyzing chronic lung diseases in a user-centered approach

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Abstract

Objectives: Medical practitioners analyze numerous types of data, often using archaic representations that do not meet their needs. Pneumologists who analyze lung function exams must often consult multiple exam records manually, making comparisons cumbersome. Such shortcomings can be addressed with interactive visualizations, but these must be designed carefully with practitioners' needs in mind.

Materials and Methods: A workshop with experts was conducted to gather user requirements and common tasks. Based on the workshop results, we iteratively designed a web-based prototype, continuously consulting experts along the way. The resulting application was evaluated in a formative study via expert interviews with 3 medical practitioners.

Results: Participants in our study were able to solve all tasks in accordance with experts' expectations and generally viewed our system positively, though there were some usability and utility issues in the initial prototype. An improved version of our system solves these issues and includes additional customization functionalities.

Discussion: The study results showed that participants were able to use our system effectively to solve domain-relevant tasks, even though some shortcomings could be observed. Using a different framework with more fine-grained control over interactions and visual elements, we implemented design changes in an improved version of our prototype that needs to be evaluated in future work.

Conclusion: Employing a user-centered design approach, we developed a visual analytics system for lung function data that allows medical practitioners to more easily analyze the progression of several key parameters over time.

Key words: visual analytics; lung function tests; qualitative research; time series visualization; disease progression.

Introduction

Visual analytics has found application in many different contexts, including the health sector. In the University Hospital Mannheim, the evolving landscape of healthcare demands systems capable of facilitating the visual analysis of patient data. This need is particularly pronounced in primary care, where physicians face the challenge of integrating comprehensive patient data efficiently. Existing IT infrastructure does not provide a unified view of multiple lung function tests, forcing physicians to individually locate and review each relevant document for a patient (see Figure 1 for an example document)—a process that is both time-consuming and prone to mistakes.

The interpretation of lung function test results is a complex task that requires specialized expertise. These tests involve a range of numerical scales for various parameters, each with their own reference values. Although current systems indicate such reference values and normal ranges, they do not offer a unified view that integrates these diverse data points. Furthermore, the severity classification of results is often relegated to remote sections of the system, if it is included at all. Compounding these issues is a lack of data integration from different source systems, necessitating manual retrieval of critical clinical data such as laboratory values.

In response to these challenges, this article outlines our contributions in 3 key areas: (1) the description of a user-centered design process, tailored to gather requirements and tasks specific to patient care in a pulmonology setting, alongside the iterative design of 2 visualization prototypes; (2) a formative evaluation of our initial prototype, providing insights and guiding further refinement; and (3) the introduction of a novel system designed to enhance the visual analysis of patient progress, particularly for those with chronic obstructive pulmonary diseases (COPD) like asthma. This system aims to streamline the data review process, offering a cohesive and integrated view of patient information. A partial version of our application is described in a EuroVis poster.[1]

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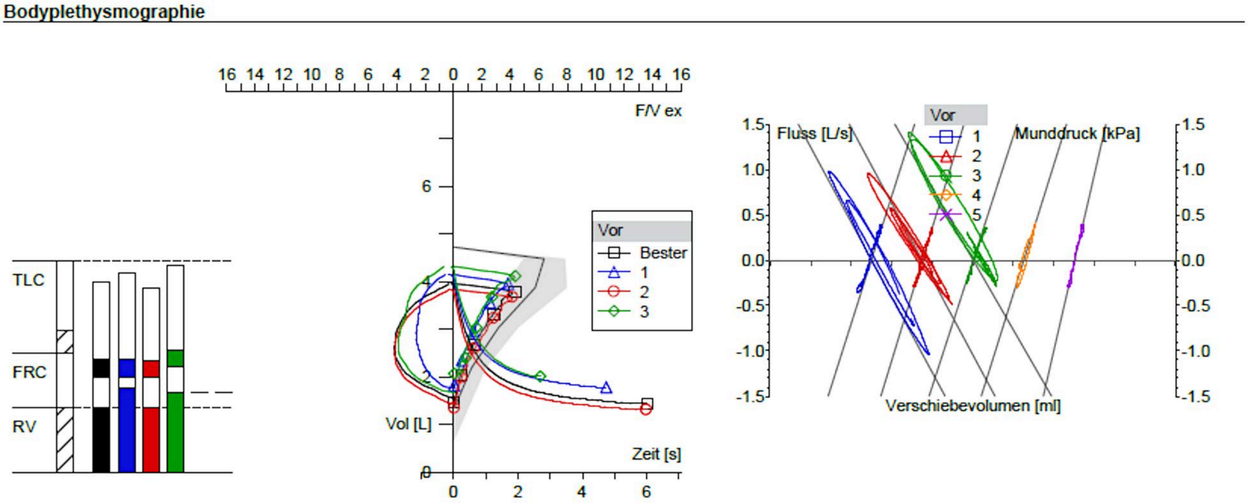

		B	%B/S	Soll	Soll uG	Z-Score	V1	V2	V3	V4	V5
Testdatum											
Testzeit											
VC MAX	L	2.65	79	3.37	2.69		2.43	2.51	2.65		
VC IN	L	2.65	85	3.10	2.41		2.43	2.43	2.65		
FVC	L	2.51	75	3.37	2.69		2.37	2.51	2.29		
FEV1	L	1.76	63	2.81	2.25		1.68	1.75	1.76		
FEV1%M	%	66.47	79	83.86	72.56		69.23	69.72	66.47		
ERV	L	0.66	58	1.14	1.14		0.24	0.66	0.53		
PEF	L/s	4.32	68	6.37	4.88		3.82	4.14	4.32		
MEF 75	L/s	2.79	49	5.72	3.50		2.52	2.79	2.68		
MEF 50	L/s	1.42	45	3.13	1.96		1.39	1.42	1.57		
MEF 25	L/s	0.51	41	1.23	0.61		0.56	0.51	0.83		
MMEF	L/s	1.20	38	3.13	1.96		1.22	1.20	1.46		
R IN	kPa/(L/s)	0.67	224	0.30	0.30		0.62	0.75	0.73	0.67	0.43
R EX	kPa/(L/s)	0.67	223	0.30	0.30		0.72	0.67	0.64	0.68	0.47
R tot	kPa/(L/s)	0.59	198	0.30	0.30		0.67	0.59	0.64	0.53	0.44
sR tot	kPa*s	1.31	136	0.96	0.96		1.48	1.31	1.42	1.16	0.97
FRCpl	L	2.02	81	2.51	1.68		2.01	2.02	2.23		
RV	L	1.36	99	1.37	0.79		1.77	1.36	1.70		
TLC	L	4.01	90	4.44	3.45		4.20	3.87	4.36		
RV%TLC	%	33.88	110	30.86	21.27		42.20	35.14	39.15		
BF	1/min	16.88	84	20.00	20.00		16.80	17.06	16.47	16.84	17.28
IC	L	1.99	101	1.96	1.96		2.19	1.85	2.12		
KVol	L	1110.00					1110.00	1110.00	1110.00	1110.00	1110.00

Figure 1. Exemplary document of a lung function test from the Thoraxklinik in Heidelberg, similar to those that medical practitioners at the University Hospital Mannheim use.

Background and significance

Software continues to impact users' interaction with medical data and patient care. The field of visualization is no exception, having produced many different applications that support domain experts by visualizing electronic health records (EHR)[2,3] or electronic medical records (EMR).[4] However, designing applications for medical practitioners can be difficult, as they often fail to capture the complexity of clinical practices, showing the importance of user-centered design approaches.[5]

When focusing on time-oriented medical data, many topics emerge: While VISITORS[6] employs domain ontologies to derive meaningful abstractions for groups of patients and TIMESPAN[7] visualizes treatment times for multiple stroke patients, other works look at the geographic distribution of various diseases.[8–10] Although aggregation can be employed to make larger collections of data tractable and easy to understand,[11,12] Caballero et al.[13] visualize hospital event logs to let users explore clinical pathways and patient data.

For the visualization of time-oriented medical data of single patients, many approaches use timelines as they were introduced by software like LIFELINES.[14] When time is of the essence, eg, for those working in Intensive or Emergency Care Units, some tools allow users to keep track of vital signs or other critical data efficiently.[15–18] Similar to our scenario, some visualizations focus on the analysis of the long-term progression of a specific disease or state of a patient.[19–22] In particular, Pieczkiewicz et al.[23] use multiple line charts to visualize lung function data after a lung transplantation was performed. In contrast to our application, they focus on data obtained over a shorter period of time that contains a larger number of data points. Additionally, our system displays reversibility testing data to let practitioners assess the effectiveness of medical treatments. Reversibility tests denote the process of taking several lung function exams with a spirometer before and after a medication has been administered. Such tests allow medical practitioners to observe potential changes that result from a medication.

For more in-depth overviews of available literature, we refer to Rind et al.,[24] Wang et al.,[25] Rostamzadeh et al.,[26,27] and Blandford[5] or the scoping review by Scheer et al.,[28] who surveyed visualizations for the comparison of individual patients to other patients or their cohorts.

Materials and methods

This section describes both the design process and evaluation of our developed visual analytics system that aims to improve patient care and reduce the workload practitioners face in clinical settings.

System design

We employed a user-centered design (UCD) approach[29] to understand and incorporate physicians' needs into the design process, as is common for problem-driven visualization research that supports domain experts. During collaboration, practitioners provide the necessary insight into domain problems and workflows that visualization experts can translate into appropriate designs for visualizations and interactions. To start off, we performed a requirement analysis in a collaborative workshop. Taking the workshop results as our foundation, we developed a prototype system that integrates feedback from regular expert consultations.

Expert workshop

In collaboration with 5 pneumologists and 2 visualization experts, we collected requirements and common tasks in a 2-hour workshop. The workshop was conducted remotely as a video conference due to the Coronavirus pandemic and used a whiteboard system to allow participants to collaborate. After introducing the interface, we employed a step-by-step workflow. First, the domain experts were tasked with filling out post-it notes that describe common tasks and routinely asked questions that arise in their day-to-day working activities. To aid participants during this process, we provided 3 guiding questions:

- What do you want to *know*?
- What do you want to *do*?
- What do you want to *see*?

Participants organized the resulting post-it notes into 3 categories: patient care, research, or both (see Figure 2). Each expert had to vote for the most relevant notes based on their experience and needs. Afterwards, 2 groups with 1 visualization expert each were formed, which then chose 2 of the highest voted notes and fitting data from available real-world datasets. Finally, the groups co-created visualization drafts based on their chosen data that should help to answer questions or perform tasks from the selected notes.

Design goals and tasks

Based on the results from the workshop, we formulated 6 high-level design goals and identified 4 tasks oriented towards patient care that served as a guideline throughout the implementation process of our system.

DG-1: Show up-to-date information about the patient (age, gender, height, weight);

DG-2: Provide an overview of all performed lung function exams over time;

DG-3: Provide an overview of all lung function parameters over time;

DG-4: Visualize the progression of each lung function parameter and enable inspection of details;

Figure 2. Screenshot of the whiteboard and post-it notes from the workshop, showing how different notes are placed in the 3 available categories.

DG-5: Visualize reversibility testing results for each lung function parameter to assess effectiveness;

DG-6: Visualize laboratory results over the time span of all lung function exams.

T1: Analyze the long-term progression of a disease;

T2: Provide a quick medical report for each lung function exam;

T3: Identify severe or problematic lung function parameters;

T4: Assess effectiveness of reversibility tests.

Data description

Data for our application were fetched from a clinical data warehouse (CDW) in Mannheim. The CDW contains preprocessed data and staged source databases from the clinical source systems Clinical win data (lung function test results) and iMed (laboratory information system). The lung function dataset was extracted from the staged source databases and additional data items from other tables of the healthcare information system. For the extraction, we set up an SQL query with the constraints described in Table 1.

The resulting dataset comprises 160 columns and contains all lung function parameters, including pre- and post-broncholytic value pairs. The initial prototype was made to accommodate up to 10 exams per patient, based on expert advice and data from the warehouse. For our system, we selected laboratory data and the most important parameters of 4 different groups based on expert feedback, which are listed in Table 2. Severity classes for the different parameters are based on established rules that can be found here: http://lungenfunktion.eu/index.htm.

Visual analytics system

As a foundation for our system, we created several sketches using the workshop results and our design goals. By frequently discussing our ideas with 1 physician, we were able to refine them to fit users' needs. The final draft was implemented as an interactive prototype and presented to workshop participants and additional experts to collect more feedback. Since usability has a high priority, we built a web-based system, which increases the application's mobility and removes the need for software installations—both of which were voiced as desirable features. In the backend, we opted for a python based flask server that uses SQL to fetch data, which is loaded, cleaned, and aggregated using pandas.[30] To create the different visualizations, we use Altair,[31] which is based on Vega Lite.[32] Its simplicity allows for rapid prototyping to quickly test and modify design ideas. For the first prototype, we prioritized those features over the adaptability of low-level libraries like D3.[33] The complete system consists of multiple connected components that allow users to interpret and analyze medical exams and lung function parameters over time. Figure 3 shows an exemplary screenshot of our prototype, and the following paragraphs give in-depth descriptions of its different components and their functionalities.

Table 1. The data features and related criteria that were used to query the staged source databases.

Data feature	Constraints
Time	Between January 1, 2018 and June 31, 2021
Diagnosis	Main or secondary ICD-10 diagnosis key corresponds to J44
Hospital department	Patient had an encounter at a ward associated with a specific department code (inner medicine, gastro, etc)
Lung function dataset	At least 1 record in the lung function database table exists for the patient

Patient details

At the top of the page, a header lets users select a patient and see related information about them (**DG-1**). Patient selection is performed using a dropdown widget and the patient information, such as age and gender, is displayed right next to the dropdown (see Figure 3A).

Overview charts

In accordance with **DG-3**, the overview charts provide a holistic view of all available measurements. The measurements are visualized as small multiples[34] in the form of area charts (see Figure 3B). Due to their small size, all measurements can be seen at a glance. Since no specific values can be read from these charts, the last measured value is displayed as text next to each chart.

Timeline

To provide information about the temporal dimension of the data and related medical records (**DG-2**, **T2**), we created a timeline component with a grey circle for each performed lung function exam (see Figure 3C). Hovering over a circle shows the text of the medical report findings in a tooltip.

Parameter charts

In line with design objectives **DG-4**, **DG-6**, and task **T1**, the main component displays a line chart for each selected parameter (see Figure 3D). The charts are arranged vertically and show the complete progression of a parameter over time. Data points are displayed as colored circles connected by a line. Different measurements can be analyzed in detail through interactions like zooming and panning. All line charts are grouped into category sections, ranging from spirometry to blood gas values and laboratory data (see Table 2). In case multiple data points are associated with one exam, the displayed circle represents the last value (for example *after* bronchodilation). To view detailed information, users can hover over a data point which yields the absolute value and, if available, the percentage of the recommended value in a tooltip. Parameters that have associated severity categories have the background of their line chart color-coded to show

Table 2. The different categories of lung functions parameters and laboratory data employed in our system.

Category	Parameters
Spirometry	FEV_1, FVC_{ex}, IC, FEV/IC_1
Body plethysmography	RV, RV/TLC, $sRAW_{tot}$
Transfer factor	DLCOSB, K_{CO}
Blood gas	pH, PAO_2, $PACO_2$
Laboratory data	Eosinophiles, Immunoglobulin (in serum)

Abbreviations: DLCOSB, single-breath carbon monoxide diffusing capacity; FEV_1, forced expiratory volume over 1 second; FVC_{ex}, forced expiratory vital capacity; IC, inspiratory capacity; K_{CO}, Krogh factor; $PACO_2$, partial pressure of carbon dioxide; PAO_2, partial pressure of oxygen; pH, potentia Hydrogenii; RV, residual volume; $sRAW_{tot}$, specific total airway resistance; TLC, total lung capacity.

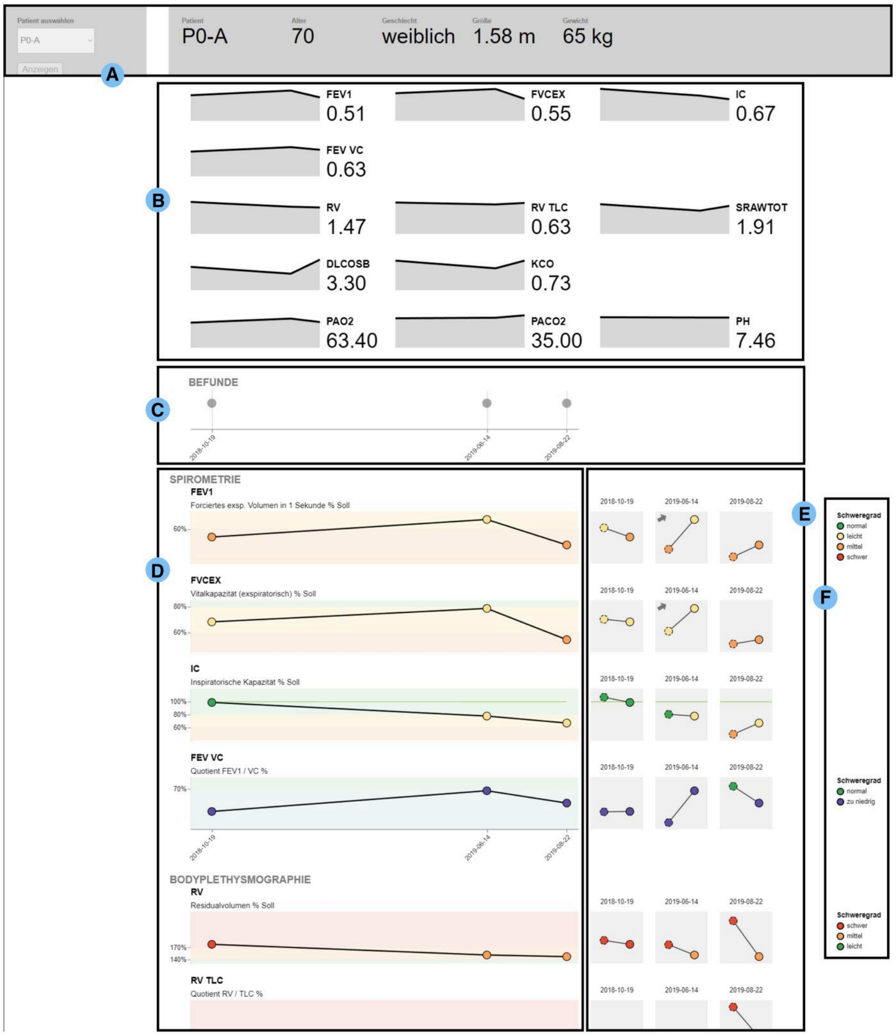

Figure 3. An exemplary view of our initial prototype, showing (A) the patient details, (B) overview charts for all lung function parameters, (C) a timeline of the patient's medical record, (D) detailed line charts for each lung function parameter ordered by category, (E) slope charts visualizing reversibility test values, and (F) severity classes for different parameters.

the varying categories (eg, *reduced*, *normal*, and *elevated*) based on established classifications (see Figure 3F).

Slope charts

The final component focuses on reversibility tests. An exam can contain 2 values, for example before (pre) and after (post) administration of a medication. Since the parameter chart only presents the post value, but does not visualize changes between values associated with an exam, slope charts were added (**DG-5**, **T4**). They are positioned on the right side of their respective parameter chart and allow users to inspect and analyze the changes resulting from a treatment (see Figure 3E). While a single slope chart represents the bronchospasmolytic test at the given date, multiple slope charts let experts examine the results over time. Each chart directly encodes the relative improvement or deterioration through the angle between the pre- and post-value circles, which are connected with a line. Circle colors indicate the measured severity using the same color scale as before and border styles are varied to distinguish the pre (dashed) and post (solid) values. As a response to **T3** and based on published medical guidelines, we chose to include an additional visual indicator for the FEV_1 and FVC_{ex} parameters, an angled arrow allows for quick identification of broncholysis success. Additionally, the slope charts support 2 hover interactions: Hovering over the slope chart background shows a tooltip that displays the value difference between before and after bronchodilation. Hovering over a circle shows details for the measured parameter in a tooltip.

Evaluation

We conducted a formative study using expert interviews to evaluate the usability of our developed prototype. To that end, we remotely interviewed 4 male pneumologists who work at different hospitals across Germany via Zoom. Each session lasted 30 minutes, was recorded on video and employed the think-aloud method[35] to collect participants' thoughts. This approach removed the need for travel, which would have severely complicated finding participants. Questions for the interview were designed in collaboration with an expert for the assessment of lung function data, who also provided correct answers for each question. Our aim was to test common scenarios that might arise during the day-to-day working activities of our target group. The same expert also took part in the selection process of the patient data used for the evaluation. All final questions and answers were additionally validated by a second expert.

Table 3. Selection of translated questions with varying degree of difficulty used during the interviews of the study.

Difficulty	Question	Goal
Low	What is the FEV_1 value of the patient for his last medical exam?	Find and read values using the visualization.
Medium	How did the obstruction develop over time?	Comparing multiple values and degrees of severity over time.
High	Did the findings improve in between time point A and time point B?	Interpretation of all available data.

Study procedure

At the start of each session, the interviewer introduced the study and demonstrated the main features of the prototype, after which participants familiarized themselves with the application on their own. Next, participants had to complete 2 tasks, with a different dataset for both, consisting of 6 questions each. In the first task, these questions had low to medium difficulty while the second task also included high difficulty questions. Lastly, there was room for feedback and participants were asked to fill out the system usability score (SUS)[36] questionnaire.

To curate data for the study, a domain expert selected different data for each task from a set of real-world patient records, all of which represent typical cases but with different characteristics. Before importing the data into our system, records were anonymized to maintain patient privacy. In collaboration with said expert, a question catalog was designed which closely mimics tasks that practitioners encounter in their everyday working activities. Each question was assigned a difficulty label, a degree of importance and a solution—example questions can be found in Table 3 while the complete catalog is available as supplementary material. Another domain expert reviewed the question catalog to affirm both its relevance and validity. Moreover, an external usability expert reviewed and approved our proposed study procedure. After all of the interviews were conducted, the recordings were manually transcribed and analyzed in-depth by one of the authors, eg, looking for common remarks and grouping them by content. One interview (P4) was discarded due to technical problems with the recording.

Results

All interviews were analyzed for usability feedback and participants' answers to the 12 task questions were compared to the solutions proposed by experts. In the following, we give a detailed account of our analysis results.

Evaluation results

Analysis of the interview recordings showed that participants largely gave valid answers that either matched experts' expectation or were open for debate. All participants came to the same conclusion for 9 out of 12 questions and only expressed minor differences for the *high* difficulty questions that required interpretation. Overall feedback was mostly positive, with participants emphasizing the easy comparison between exams and the fast access to data. This is well-illustrated by a comment participant P3 made: "*For the tool itself, I find it relatively clear at first glance to see the value progression over time. This is (...) actually made quite clear by the curve (...) with the color coding.*" Two participants explicitly mentioned the time-oriented design as a useful feature, whereas a third participant favorably noted the practicality for everyday life. Moreover, the scores from the SUS questionnaires indicate that participants found the system usable, with 3 out of 4 participants rating it with high values over 85 (P1: 87.5, P3: 92.5, P4[1]: 95) and 1 participant rating it slightly below 80 (P2: 77.5).

With regard to limitations, multiple opportunities for improvements were mentioned or could be extracted from the recordings. These were categorized by topic and are described in the following paragraphs, supplemented by related quotes from participants.

I1: Identifying the date of a measurement

Even though the visualization always displays a time-axis, some users struggled to identify the date of a data point on several occasions. This led to redundant interactions with the visualization and made comparing multiple data points with a specific date more difficult, since only 1 tooltip is visible at a time when hovering.

Participant P3 remark: "*(...) sometimes I needed to take a look, where exactly which date can be found, (...)*"

Original: "*(...) ich musste immer so'n bisschen gucken, wo genau welches Datum jetzt ist, (...)*"

I2: Overview chart usage

While the overview charts are very prominent in the application, they were barely used by participants, if at all. This could be related to the questions we asked, since none of them specifically targeted the use of this component.

Participant P2 remark: "*And secondly I was not sure, for what exactly the curves at the top—in this case the number 0.45 FEV_1—how it is correlated to the measurements.*"

Original: "*Und als zweites mir war nicht ganz klar, wofür diese Kurven hier oben—also was die Ziffer 0.45 FEV_1—in welcher Korrelation die hier zu den Messwerten steht.*"

I3: High need for vertical space

Due to the parameter charts being stacked vertically and the large amount of information being displayed, it was not always possible to see a specific pair of charts at the same time. Therefore, users were required to scroll the page, which made it more difficult to compare values or connect information.

Participant P1 remark: "*(...) now I need to shortly try to get both on one page (...)*"

Original: "*(...) jetzt muss ich einmal ganz kurz ein bisschen hier beides auf eine Seite kriegen (...)*"

I4: Data-dependent horizontal width

Another issue was the variable width, which depended on the number of exams contained in the datasets. Since each exam creates an associated slope chart with a fixed size that is displayed next to the parameter chart, the needed width depends on the number of exams. The visualization may therefore not

[1] Only the SUS questionnaire is available for this participant. The interview was discarded due to technical difficulties.

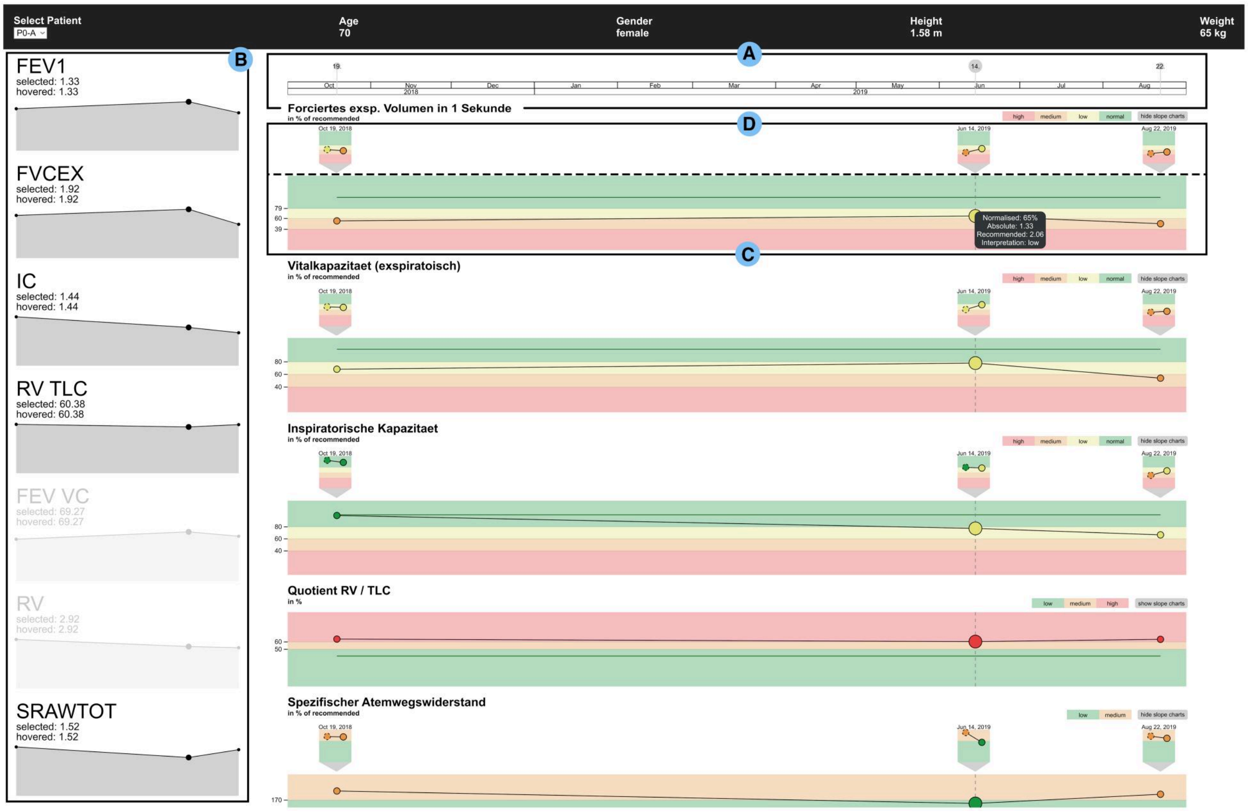

Figure 4. View of the revised system with improved (A) timeline, (B) overview charts, (C) parameter charts, and (D) slope charts. In this example, 2 parameter charts and the slope charts for RV/TLC are hidden, but they can be shown again on demand. A limited version of the revised system with hard-coded data is available at https://rwarnking.github.io/lufuvis-light.

fill the complete width or may not fit without horizontal scrolling. Both conditions are inconvenient and can make comparison cumbersome.

Design improvements

We re-evaluated several design choices and devised multiple improvements in order to address issues found during the evaluation. These improvements required a technological reorientation, as Altair was not capable of the functionalities our improvements required. As a consequence, we switched to D3 for the implementation of the visualization frontend. Figure 4 shows an exemplary view of the improved system. In the following, we describe the most important changes and how they relate to the issues identified in the previous subsection.

Timeline and parameter charts

The timeline is now displayed as a sticky header (see Figure 4A), resulting in a fixed point when searching for a related date (**I1**). In addition, clicking on a data point in a line chart permanently marks it as selected, visually indicated by increasing its radius across all charts. Hovering over any data point in the timeline displays a dashed indicator line in *all* parameter charts (see Figure 4C). Finally, the timeline is synchronized with the parameter charts' zoom transform.

Overview charts

We opted to move this component from the top to the left of the application as a sticky sidebar with a fixed width to conserve vertical space (**I3**). In addition, we gave these charts new functionalities (**I2**). Clicking on any of the charts allows users to toggle between showing or hiding the associated parameter chart (see Figure 4B). Furthermore, users can now reorder the parameter charts by dragging the corresponding overview chart to a different position. These changes let users control *which data* is shown at *which location* in the application, making direct comparison easy and letting them hide unneeded data. If a data point is clicked on or hovered over in a line chart, the related values are displayed above each overview chart and corresponding circles are increased in size.

Slope charts

To address issue **I4** and improve the usability of the slope charts, we explored several new approaches to visualize the reversibility testing results. The new options we considered in more detail were *layering* and *repositioning*.

Layering incorporates the data directly into the line chart, similar to candlestick charts. This removes the slope charts altogether, resulting in a more compact visualization, but also creates overdraw and visual clutter. For example, minor changes might not be visible due to overdraw, which can result in misinterpretation by the user. Figure 5 illustrates different variations that we considered for this approach.

The second option was to reposition the slope charts above each parameter chart, which allows the latter to occupy more screen width. Furthermore, it connects each slope chart directly to its corresponding time point, displaying it at a similar horizontal position. This approach was initially discarded

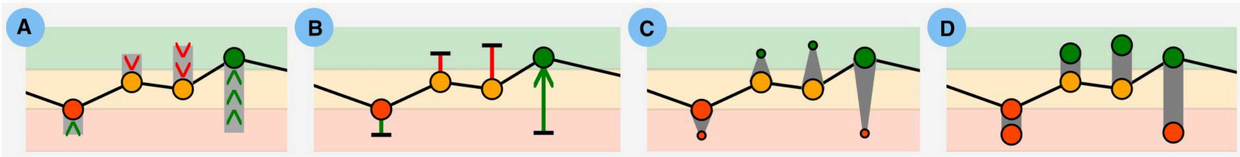

Figure 5. Different options we explored for the layering approach to revise the slope charts. They visualize the changes from pre- to post-medication values using (A) bars and arrow tips, (B) arrows, (C) circles connected via triangles, and (D) band-connected circles.

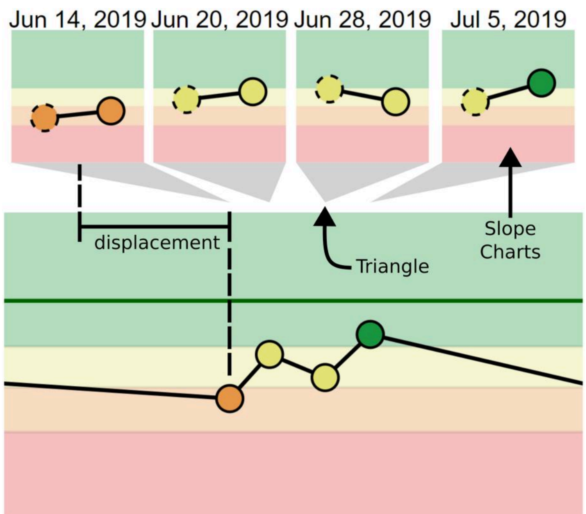

Figure 6. Zoomed in view of an example where the force-directed positioning of the slope charts prevents them from overlapping. Gray triangles point to the correct horizontal position for each chart.

due to limitations of the Altair framework and concerns regarding overdraw between charts. Switching to D3 solved both problems and allowed for displacement of the slope charts in a force-directed manner. Each slope chart is first positioned at the exact horizontal location of its related data point. Then, a force-directed simulation is run, similar to a force-directed graph layout.[37] Neighboring slope charts repel each other, while being pulled towards their correct horizontal position. In this manner, the charts deviate as little as possible from their original position but also avoid overlap, as illustrated in Figure 6. To emphasize its exact point in time, each chart is connected to its related data point using a triangular indicator that points to the correct horizontal position. When users zoom or pan in the parameter charts, the system adjusts the positions of all slope charts and re-runs the simulation. After exploring both options, we settled on repositioning the slope charts, since it does not drastically modify the line charts and more directly addresses participants' critiques. While this choice increases the need for vertical space, this is counteracted by allowing users to hide the slope charts on demand.

Discussion

Practitioners could comfortably use our prototype to solve tasks of varying difficulty, which were designed to closely resemble their day-to-day working activities related to lung function data. They generally liked the system and had little trouble navigating it, even with limited training. However, participants' interactions with the system uncovered some usability and utility issues that we addressed in an improved iteration of our prototype. Most notably, the overview charts were repurposed to give users more freedom in arranging visualizations in the application. Additionally, the slope charts were repositioned to strengthen the connection to their related exam. Several modification were made to improve the general layout and space requirements. How these changes affect user experience needs to be investigated in more detail in future work. Such an evaluation could also provide the opportunity to test our system against standard tools currently employed by practitioners or to investigate the different slope chart designs. The user-centered design approach we employed helped us create a system that addresses users' real needs and could be used in other contexts where required knowledge is distributed across different experts. Finally, we can envision more usage scenarios for our application, eg, to communicate findings to patients directly or to help medical students learn about lung function data analysis.

Conclusion

Collaborating with medical practitioners, we designed and implemented an interactive visualization system that supports medical experts who analyze lung function data across multiple medical exams. We employ connected line charts and incorporate domain knowledge in the form of parameter-dependent severity categories, offering a cohesive and integrated view of the patient's disease progression.

Our formative study showed that the system provides easy access to relevant data and that it supports temporal data analysis, which could help practitioners in their day-to-day work activities. Participants of the study liked using our system and correctly answered domain-relevant questions with varying levels of difficulty. The study results additionally uncovered several areas for improvement, such as inconvenient interactions for the comparison of lung function parameters that are far apart in the layout. Consequently, we redesigned several components to give users greater flexibility, thereby making comparisons effortless. Based on these overall encouraging results, we aim to evaluate our redesigned system to study its effectiveness in greater detail and compare it to existing approaches employed by medical practitioners.

Author contributions

The authors confirm contribution to the paper as follows: (i) Study and workshop conception and design: Jan Scheer, Frederik Trinkmann, Till Nagel; (ii) Provision of study and workshop materials: Frederik Trinkmann, Fabian Siegel, Jan Scheer, Till Nagel; (iii) Study and workshop execution and analysis: René Pascal Warnking, Jan Scheer, Till Nagel; (iv)

Software design and implementation: René Pascal Warnking, Jan Scheer, Till Nagel, Franziska Becker; (v) Draft manuscript preparation: René Pascal Warnking, Jan Scheer, Till Nagel, Franziska Becker; (vi) Administrative support: Fabian Siegel, Till Nagel. All authors reviewed the results and approved the final version of the manuscript.

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

The authors have no competing interests to declare.

Data availability

The source code for a reduced version of our final prototype with hard-coded data is available under an Open-Source license at https://github.com/rwarnking/lufuvis-light.

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